EXISTENCE OF A LEFT ADJOINT TO THE PULLBACK OF A MORPHISM OF TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The Yoneda lemma. Categories, Lemma categories-lemma-yoneda. Shows that the representing object in the hypothesis is unique up to unique isomorphism and provides the identities used throughout.
- Sheaves as coequalizers of coproducts of representables. Same chapter, Lemma lemma-sheaf-coequalizer-representable. Produces the coequalizer presentation of an arbitrary sheaf by coproducts of representable sheaves, using that the chosen family of objects covers.

Source reference. Original lemma 001P: sources/stacks-sites/pinned-original-source.tex, source lines 5282–5282. Navigation pointer invoked at proof line 5282; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00WS: sources/stacks-sites/pinned-original-source.tex, source lines 5336–5336. Navigation pointer invoked at proof line 5336; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$, $\mathcal{D}$ be two sites. Let $f : Sh(\mathcal{D}) \to Sh(\mathcal{C})$ be a morphism of topoi. Let $E \subset \mathrm{Ob}(\mathcal{D})$ be a subset such that*

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(1) *for $V \in E$ there exists a sheaf $\mathcal{G}$ on $\mathcal{C}$ such that $f^{-1}\mathcal{F}(V) = \mathrm{Mor}_{Sh(\mathcal{C})}(\mathcal{G}, \mathcal{F})$ functorially for $\mathcal{F}$ in $Sh(\mathcal{C})$,*
(2) *every object of $\mathcal{D}$ has a covering by objects of E.*

Then f^{-1} has a left adjoint $f_!$.

Proof. By the Yoneda lemma (Categories, Lemma 001P) the sheaf $\mathcal{G}_V$ corresponding to $V \in E$ is defined up to unique isomorphism by the formula $f^{-1}\mathcal{F}(V) = \mathrm{Mor}_{Sh(\mathcal{C})}(\mathcal{G}_V, \mathcal{F})$. Recall that $f^{-1}\mathcal{F}(V) = \mathrm{Mor}_{Sh(\mathcal{D})}(h_V^\#, f^{-1}\mathcal{F})$. Denote $i_V : h_V^\# \to f^{-1}\mathcal{G}_V$ the map corresponding to id in $\mathrm{Mor}(\mathcal{G}_V, \mathcal{G}_V)$. Functoriality in (1) implies that the bijection is given by

$$\mathrm{Mor}_{Sh(\mathcal{C})}(\mathcal{G}_V, \mathcal{F}) \to \mathrm{Mor}_{Sh(\mathcal{D})}(h_V^\#, f^{-1}\mathcal{F}), \quad \varphi \mapsto f^{-1}\varphi \circ i_V$$

For any $V_1, V_2 \in E$ there is a canonical map

$$\mathrm{Mor}_{Sh(\mathcal{D})}(h_{V_2}^\#, h_{V_1}^\#) \to \mathrm{Hom}_{Sh(\mathcal{C})}(\mathcal{G}_{V_2}, \mathcal{G}_{V_1}), \quad \varphi \mapsto f_!(\varphi)$$

which is characterized by $f^{-1}(f_!(\varphi)) \circ i_{V_2} = i_{V_1} \circ \varphi$. Note that $\varphi \mapsto f_!(\varphi)$ is compatible with composition; this can be seen directly from the characterization. Hence $h_V^\# \mapsto \mathcal{G}_V$ and $\varphi \mapsto f_!\varphi$ is a functor from the full subcategory of $Sh(\mathcal{D})$ whose objects are the $h_V^\#$.

Let J be a set and let $J \to E$, $j \mapsto V_j$ be a map. Then we have a functorial bijection

$$\mathrm{Mor}_{Sh(\mathcal{C})}(\coprod \mathcal{G}_{V_j}, \mathcal{F}) \longrightarrow \mathrm{Mor}_{Sh(\mathcal{D})}(\coprod h_{V_j}^\#, f^{-1}\mathcal{F})$$

using the product of the bijections above. Hence we can extend the functor $f_!$ to the full subcategory of $Sh(\mathcal{D})$ whose objects are coproducts of $h_V^\#$ with $V \in E$.

Given an arbitrary sheaf $\mathcal{H}$ on $\mathcal{D}$ we choose a coequalizer diagram

$$\mathcal{H}_1 \rightrightarrows \mathcal{H}_0 \longrightarrow \mathcal{H}$$

where $\mathcal{H}_i = \coprod h_{V_{i,j}}^\#$ is a coproduct with $V_{i,j} \in E$. This is possible by assumption (2), see Lemma 00WS (for those worried about set theoretical issues, note that the construction given in Lemma 00WS is canonical). Define $f_!(\mathcal{H})$ to be the sheaf on $\mathcal{C}$ which makes

$$f_!\mathcal{H}_1 \rightrightarrows f_!\mathcal{H}_0 \longrightarrow f_!\mathcal{H}$$

a coequalizer diagram. Then

$$\begin{aligned}
\mathrm{Mor}(f_!\mathcal{H}, \mathcal{F}) &= \mathrm{Equalizer}(\ \mathrm{Mor}(f_!\mathcal{H}_0, \mathcal{F}) \rightrightarrows \mathrm{Mor}(f_!\mathcal{H}_1, \mathcal{F})\ ) \\
&= \mathrm{Equalizer}(\ \mathrm{Mor}(\mathcal{H}_0, f^{-1}\mathcal{F}) \rightrightarrows \mathrm{Mor}(\mathcal{H}_1, f^{-1}\mathcal{F})\ ) \\
&= \mathrm{Hom}(\mathcal{H}, f^{-1}\mathcal{F})
\end{aligned}$$

Hence we see that we can extend $f_!$ to the whole category of sheaves on $\mathcal{D}$. □